

Nuclear Fuel Performance

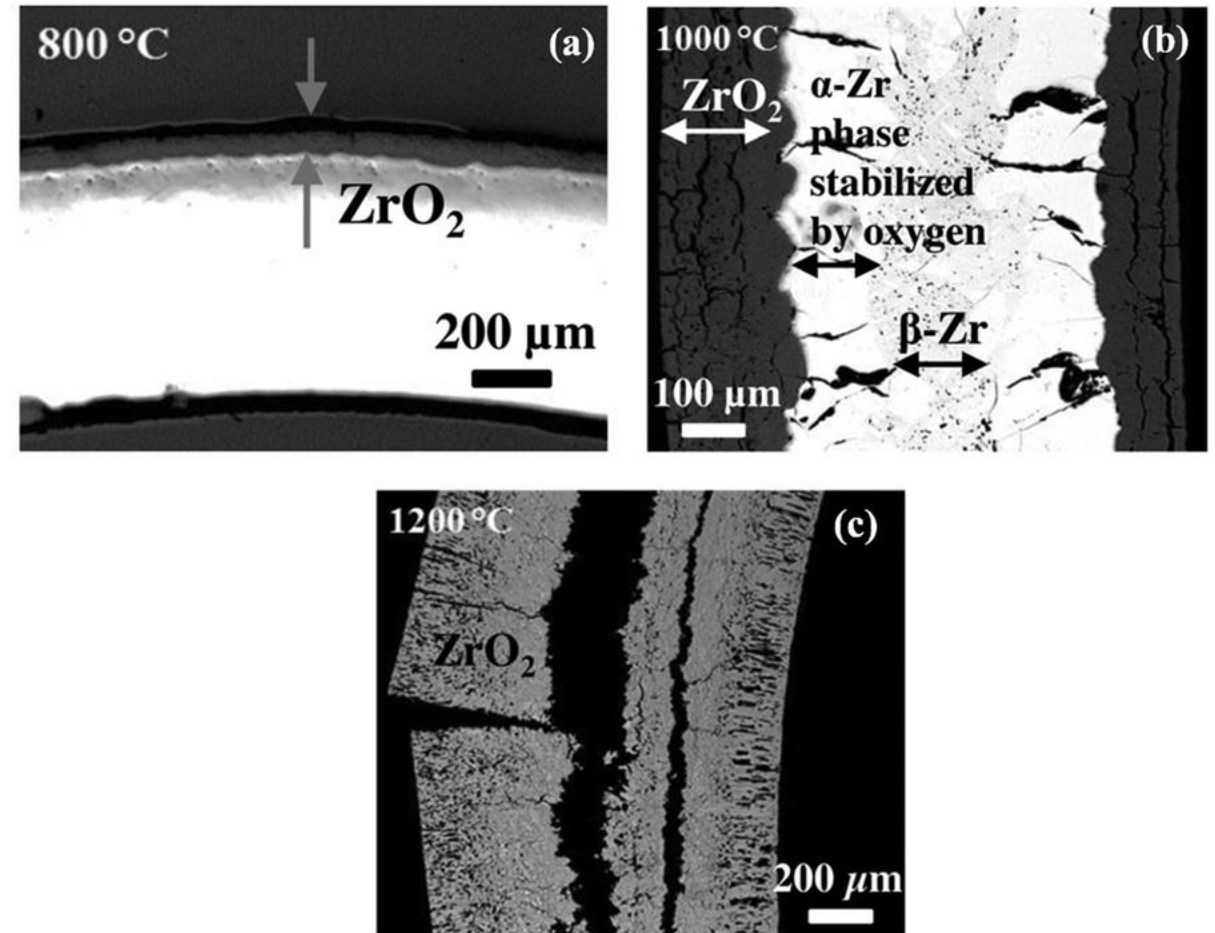
NE-533
Spring 2026

Last time

- RIA
 - control rod ejection/drop
 - fast rise in temperature/power
 - causes pellet expansion and gas bubble expansion
 - can lead to either PCMI-based failure or DNB-based failure
- LOCA
 - pipe burst
 - rise in temperature from decay heat without cooling
 - much slower than RIA
 - leads to high oxidation, high T, high hydriding, ballooning and burst
- Started on accident safety limits, talking through steam oxidation

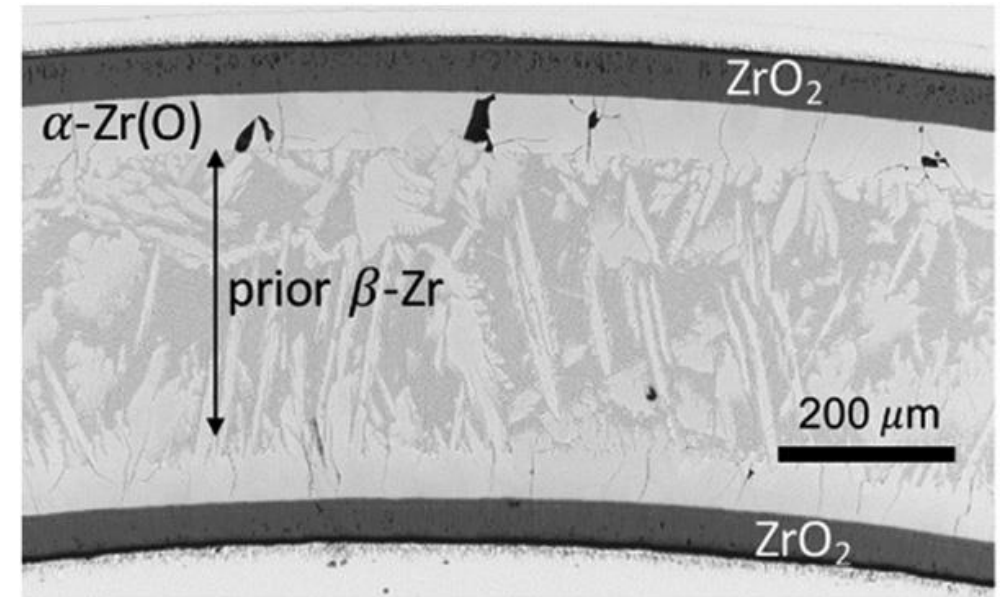
Steam Oxidation

- Right is an image of Zircaloy-2 after steam oxidation for 8 hours at a) 800 C, b) 1000 C, c) 1200 C
- At 800C, the oxide layer is intact and passivating
- At higher temperatures, the oxide layer is thicker and cracked
- At 1200C, everything has been oxidized



High T Phase Transition

- alpha Zr can phase transform to beta Zr at 863 C
- With high O content, this leads to a beta phase matrix with oxygen-stabilized alpha Zr closest to the oxide/metal interface
- This stabilized alpha phase contributes to the brittle failure behavior

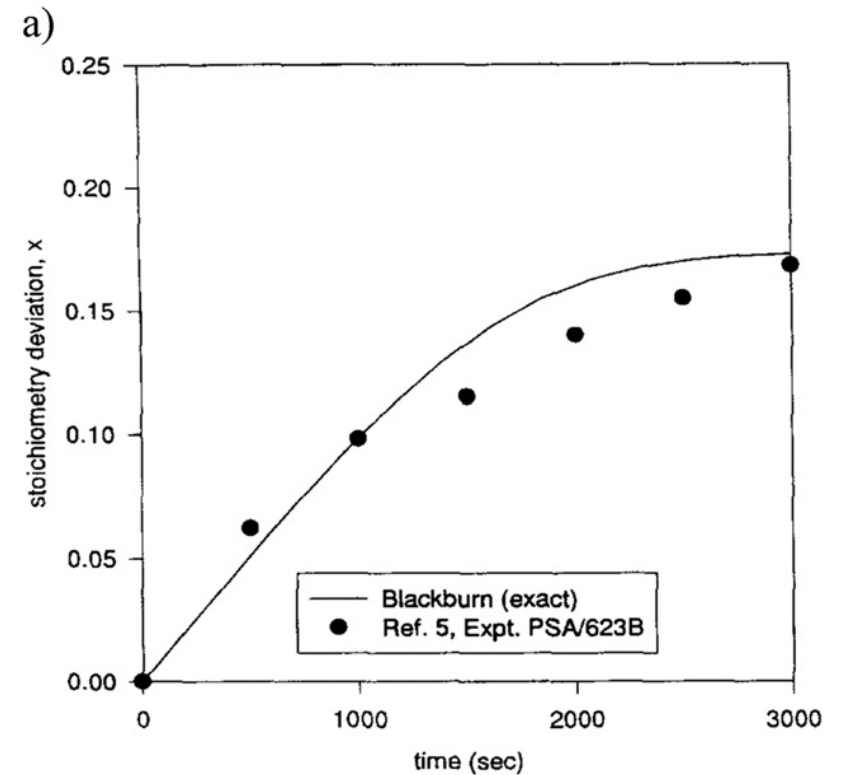


Hydride Embrittlement

- We have already covered this
- Hydrogen produced in oxidation and radiolysis can be picked up by the cladding, producing hydrides which can embrittle the cladding
- More oxidation = more hydrogen = more embrittlement

Fuel oxidation

- UO_2 is chemically inert in water, with a positive Gibbs free energy to form U_3O_8 or U_4O_9 compounds
- The oxidation of UO_2 is a dissolution reaction of water on the surface, freeing O to diffuse into UO_2 , making UO_{2+x} , and producing H in the coolant



Summary

- Steam oxidation behavior is critical for safe operation during an accident
- Breakaway oxidation can occur, leading to thick, cracked oxides and high H generation
- Zr phase transformations are indicative of high temperatures, and lead to further embrittlement of the cladding
- Fuel oxidation of UO_2 is largely not a concern, but can have deviations in stoichiometry, affecting fuel properties

ACCIDENT TOLERANT FUELS

Mitigation through design

- Accident tolerant fuels aim to provide additional coping time
 - The time required for the water to boil away
 - The time required for the fuel to melt
 - The time required for the molten fuel to breach the primary pressure boundary
- Fuels with enhanced tolerance can tolerate loss of active cooling for a considerably longer period, while maintaining or improving performance during normal operation
- Can also widen the existing safety margin for nuclear plants and improve nuclear plant performance with fuel that lasts longer

Accident Tolerant Criteria

- The safety margin is increased by reducing the fuel's centerline temperature relative to its melting temperature or another relevant failure temperature
- The release of fission products from the fuel matrix is mitigated
- Uranium density is maximized enabling increased operational flexibility by increasing cycle length, reducing enrichment, etc.
- The chemical response of the fuel is limited during exposure to the reactor coolant
- The volumetric heat capacity is minimized so that the fuel tends to transfer energy out of the pellet and into the coolant
- The potential for fuel, cladding, and coolant interaction – both chemical and mechanical – is minimized
- The impact of non-fissile fuel constituents on neutronics, fuel cycle, reactor performance, and safety characteristics is minimized

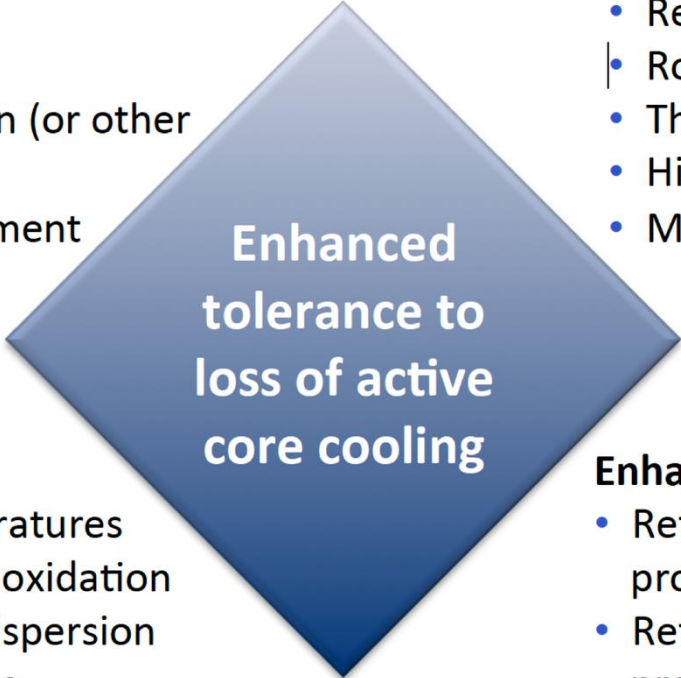
Accident Tolerance

Improved Reaction Kinetics with Steam

- Decreased heat of oxidation
- Lower oxidation rate
- Reduced hydrogen production (or other combustible gases)
- Reduced hydrogen embrittlement

Improved Fuel Properties

- Lower fuel operating temperatures
- Minimized cladding internal oxidation
- Minimized fuel relocation/dispersion
- Higher fuel melt temperature



**Enhanced
tolerance to
loss of active
core cooling**

Improved Cladding Properties

- Resilience to clad fracture
- Robust geometric stability
- Thermal shock resistance
- Higher cladding melt temperatures
- Minimizing fuel-cladding interaction

Enhanced Fission Product Retention

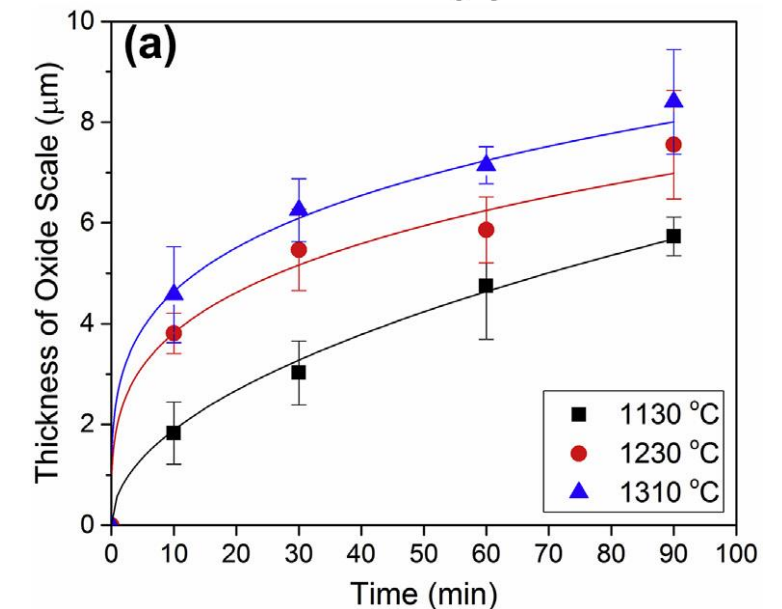
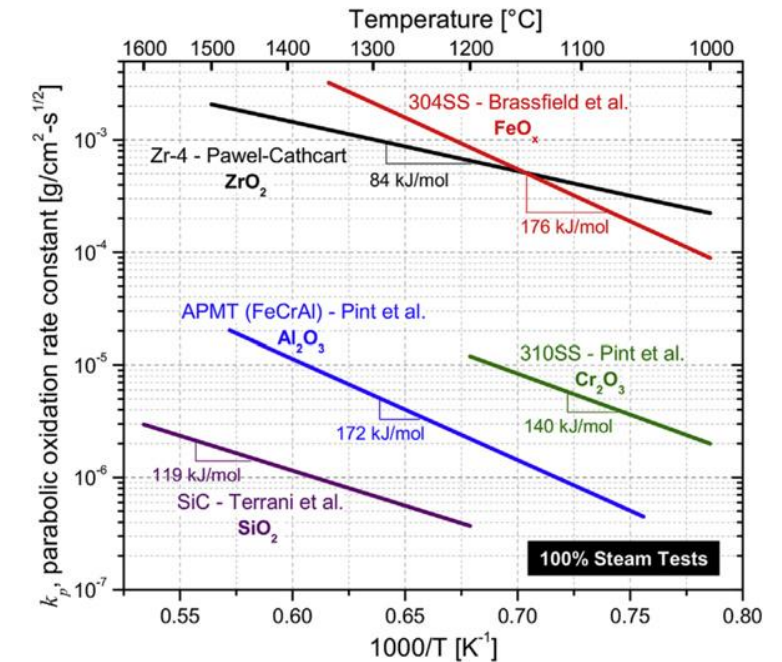
- Retention of gaseous fission products
- Retention of solid/liquid fission products

Categories of ATF Concepts

- ATF can be divided into three main categories
 - near-term ATF improvements
 - novel fuel and cladding materials
 - transformational fuel and cladding concepts
- Near-term ATF improvements
 - Improved Zr alloy cladding
 - Enhanced UO₂
 - These changes are mean to be drop-in replacements that are compatible with existing systems

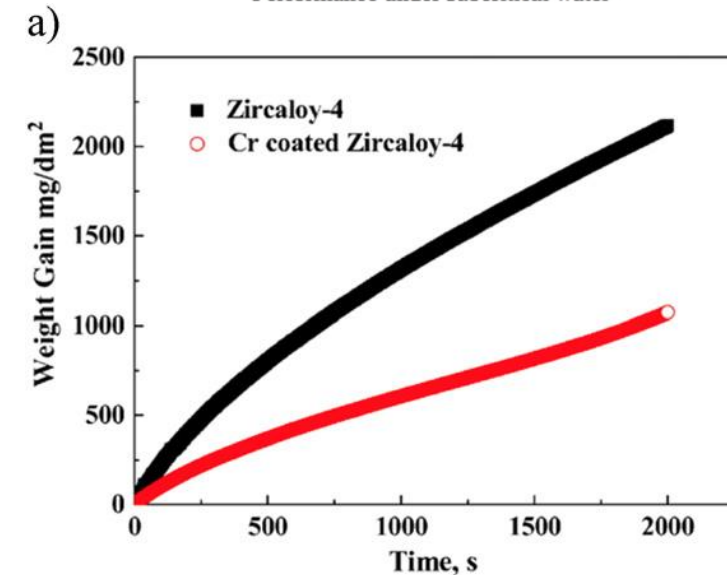
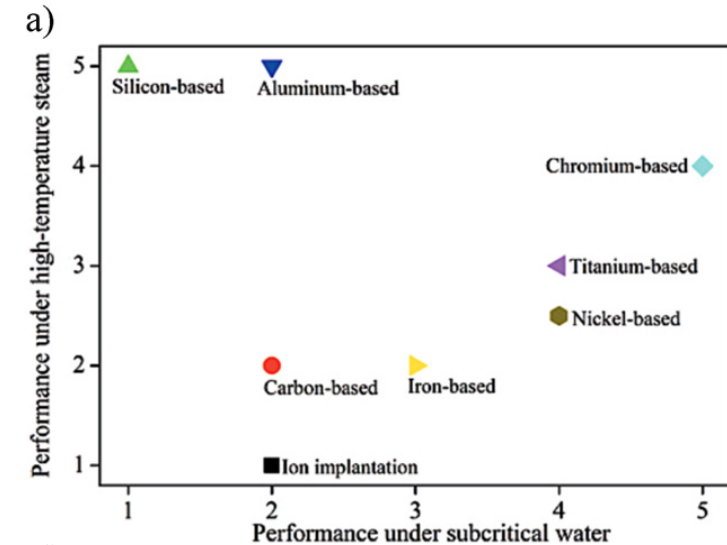
Near-term ATF Coatings

- Improved zirconium-based cladding
 - coatings to improve corrosion resistance, especially under high T steam
 - metallic coatings or ceramics nitride or MAX coatings (M=transition metal, A=group 13-16 elements, e.g., Al or Si, X=C/N)
 - typically contain either Cr, Al, or Si, to ensure a protective oxide coating
- Most widely explored are Cr metal, CrAl, and CrN coatings
- Cr coatings of 2-10 μm on Zr-4 resulted in a $\sim 50\text{X}$ reduction in oxidation rate over Zr-4 in a steam environment
- Less oxidation also yields less H, and thus less embrittlement



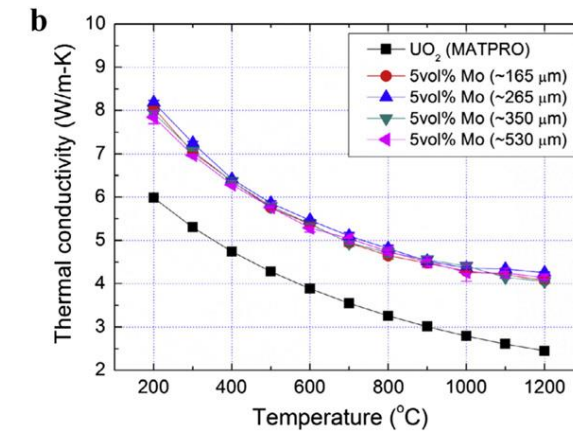
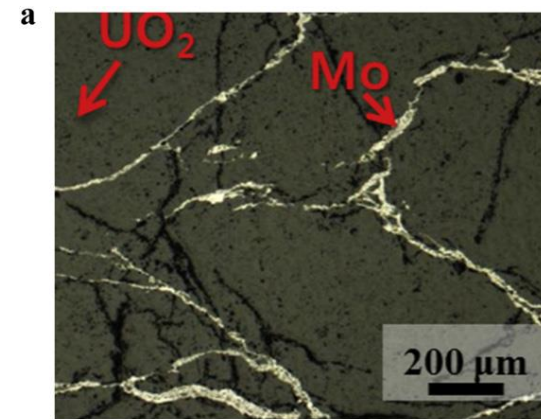
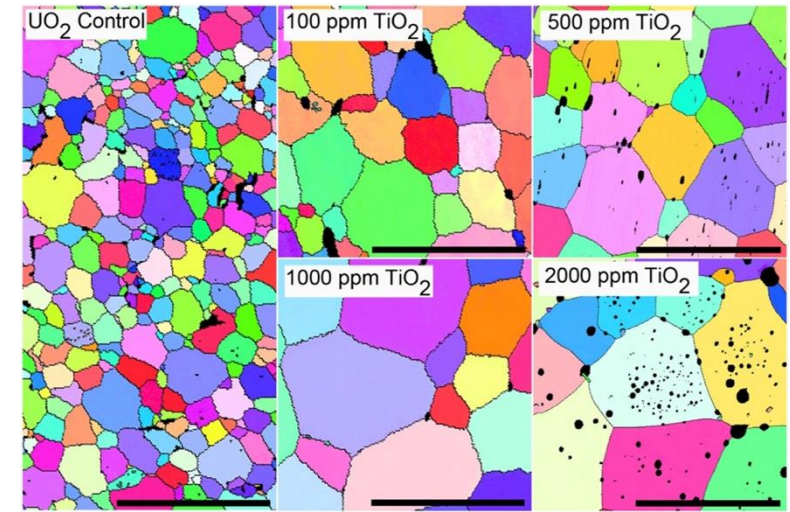
Near-term ATF Coatings

- Ceramic coatings can display excellent oxidation resistance, but can be brittle and displace CTE mismatch
- Variety of fabrication methods explored for metallic coatings
 - cold spray, CVD, PVD, sputtering, etc.
- Mechanical integrity of the coating and adherence to Zr is key
- Also have the issue of Cr-Zr eutectic reaction at 1332 C deteriorates cladding
- But, very promising and still being pursued



Near-term ATF Fuel

- Enhanced UO₂
 - minor additives to improve UO₂ properties
- Doped UO₂ typically includes a transition metal oxide (e.g., Cr₂O₃, TiO₂), that modifies the oxygen potential, subsequently modifying the oxygen kinetics, changing the grain size during sintering
- Larger grain sizes mean slower/delayed fission gas release (think of the Booth model)
- Inert secondary inclusions (e.g., Mo, SiC, diamond, BeO) involve higher levels of additives targeted to increase the thermal conductivity
- Natural development in UO₂ fuel, will happen to some degree to gain margins in safety, fuel performance, and economics

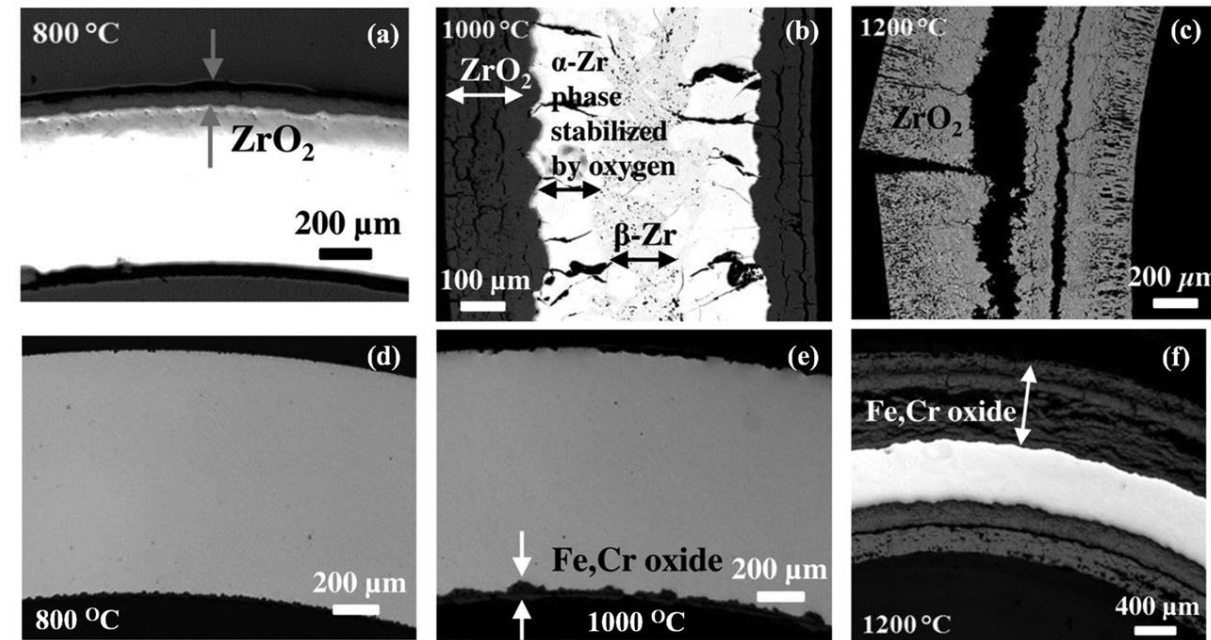


Categories of ATF Concepts

- Novel fuel and cladding materials
 - U_3Si_2 or UN fuels
 - FeCrAl alloys for cladding
 - Requires additional licensing
 - Minimal changes needed for LWR designs, could be deployed in 10ish years
- Transformational fuel and cladding concepts
 - fully ceramic micro-encapsulated (FCM) fuels, e.g., TRISO
 - SiC/SiC composite cladding
 - drastically differ from traditional LWR fuel systems
 - may require significant modifications to reactor design and operation, but offer potentially large benefits to performance and/or safety

Advanced Cladding Materials

- Fe-Cr or Fe-Cr-Al alloys display significantly improved oxidation resistance compared to Zr alloys
- Fe-Cr can still form oxides which are not protective at 1200 C
- Fe-Cr-Al alloys form an Al_2O_3 layer with enhanced stability compared to ZrO_2 , FeO_2 , or Cr_2O_3
- SiC cladding can display even better oxidation resistance



Summary

- Accident tolerant fuel includes different fuel and cladding materials to increase time before catastrophic behavior during an accident
- Aim to add coping time, increase safety margin, improve behaviors
- Four key areas:
 - improve reaction kinetics with steam, improved cladding properties, improved fuel properties, enhanced fission product retention
- Near term concepts
 - cladding coatings (Cr, CrAl, etc.)
 - enhanced UO₂ (dopants, inert additives)
- Novel fuel and cladding concepts
- Transformational concepts

LIMITING PHENOMENA

Limiting Phenomena

- Concerning limiting phenomena, some criteria have been established for the UO₂ and MOX fuel designs
- Engineering must demonstrate that all relevant parameters fulfill those criteria at any time from the loading of the fuel to reprocessing, or during long-term storage
- The key performance limiting phenomena are Pellet-clad mechanical interaction; Cladding elongation and assembly bow; Cladding oxidation and hydrogen pickup; Cladding wear; Power to melt; Fuel rod internal pressure; Departure from nucleate boiling; Normal operation limits

Limiting Phenomena

- PCMI
- PCMI is a complex process with a maximum risk for failure when the fuel pellet to cladding gap closes firmly and the reactivity of the fuel is still high
- The risk is enhanced by pellet fragments inducing a local shear strain on the cladding, and by the chemical interaction kinetics at the interface
- In order to prevent SCC, the cladding hoop stress calculated for normal operation and transients is limited
- The extent of the total permanent hoop strain is limited during the whole lifetime of the fuel rods, typically to 1%

Limiting Phenomena

- Cladding elongation and assembly bow
- During irradiation, the anisotropic character of the cladding material and the preferential migration of vacancies and interstitials in specific lattice planes drive an overall cladding axial growth, activated by the fast neutron flux
- When contact is established between the pellet and the cladding, pellet axial elongation causes an additional axial cladding strain
- This can lead to fuel rod bow with pitch reduction between the rods, reducing thermal margins
- Differential elongation of guide tubes in a PWR assembly can lead to an overall assembly bow

Limiting Phenomena

- Cladding oxidation and hydrogen pickup
- For the ZrO_2 formation at the cladding waterside surface, a typical criterion is related to the ASTM criterion of a maximum cladding wall thickness reduction of 10%, which corresponds to an oxide thickness of the order of 100 microns
- When the hydrogen concentration in the cladding exceeds the solubility limit, 70–100 ppm by weight at operating temperatures, zirconium hydrides will form
- The impact of hydrides on key mechanical properties depends strongly on hydride distribution and orientation
- Oxidation and hydrogen pickup are increasingly important at higher exposures, as the dependence on burnup is nonlinear

Limiting Phenomena

- Cladding wear
- The criterion for cladding wear at the contact points between grid spring/dimples and the fuel rod is often also related to the ASTM criterion of a maximum cladding wall thickness reduction of 10%
- More wear is technically acceptable, as evidenced from operational experience

Limiting Phenomena

- Power to melt
- The use of uranium dioxide or MOX provides a comfortable power to melt margin
- The melting temperature decreases slightly with burnup, but remains above 2750C
- At high burnup, above 50 MWd/kgU, considering the fuel thermal conductivity decrease, the power to melt is estimated to be around 600 W/cm, which is an unrealistic high LHR

Limiting Phenomena

- Fuel rod internal pressure
- Significant reopening of the radial gap between the fuel stack and the cladding must be avoided to ensure the heat transfer to the coolant
- If a gap opens, fuel overheating and excessive fission gas release can occur, ultimately leading to fuel failure
- The original criterion required that the rod inner pressure must never exceed the outer coolant pressure
- This criterion was over-conservative and has been replaced by a 'nonlift-off' criterion, where the radial creep-out of the cladding (driven by gas pressure in excess of the system pressure) must never exceed the expansion rate of the pellet
- Experiments have shown that a large overpressure of the gas (considerably more than 5.0 MPa) is needed to initiate the reopening

Limiting Phenomena

- Departure from nucleate boiling
- With increasing heat flux there comes a point at which the heat transfer from a fuel rod rapidly decreases due to the insulating effect of a steam blanket that forms on the rod surface, resulting in a severe increase of cladding temperature and possibly cladding failure
- The ratio of the heat flux needed to cause departure from nucleate boiling (DNB) at given local coolant properties (pressure, enthalpy, mass flow rate) to the actual local heat flux of a fuel rod is defined as the DNBR
- This phenomenon may limit the maximum allowed thermal power of a given PWR

Limiting Phenomena

- Normal operation limits
- Constraints on the axial LHR distribution are typically applied at the core design level and during normal operation to guarantee that the conditions are never worse than those assumed in scenarios considered in the accident analyses
- The maximum allowed LHR may depend on the axial position and on burnup, and can be reactor- and even cycle-specific
- The fulfilment of the constraint is verified during the reload safety evaluation process as well as during plant operation

Summary

- There are a variety of limiting phenomena in LWR fuel systems that provide the boundaries of operation and lifetime
- These limits include phenomena in the fuel, gap, cladding, corrosion, and assembly levels
- The performance of uranium dioxide and MOX fuels in LWR nuclear reactors is well established
- These fuels have demonstrated a very good behavior during irradiation, favored by their high melting temperature, providing large operating temperature margins